Smart Emergency Traffic Light Control System

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Project Proposal Report

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DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Traffic congestion at city intersections is a major problem, especially when traffic lights cannot see the actual number of vehicles on the road. This problem becomes worse when emergency vehicles need a clear path, because slow traffic lights can delay the entire emergency response. In Sri Lanka, roads carry most of the passengers and goods, making smart traffic control very important for daily travel and public safety.

The main problem with existing systems is that they usually focus only on changing traffic light times or only on finding emergency vehicles. Many systems also use very expensive hardware and only work well in clear daylight. Also, most solutions do not have a practical, camera-based system that can monitor all four directions of a junction clearly at night or in bad weather.

To solve this problem, this research proposes a smart camera system to detect traffic density at a four-way junction. The system uses four standard cameras (one for each road) to record live traffic. A Raspberry Pi mini-computer processes this video using AI to identify and count vehicles, and to calculate how crowded the road is. To solve the problem of dark nights and bad weather, the system uses a special AI-based thermal support method (night vision) to see vehicles clearly without using expensive thermal cameras. After calculating the traffic density, this data is sent to the cloud to help the main Smart Emergency Traffic Light System make decisions.

The technologies used include standard cameras, Raspberry Pi, computer vision AI, cloud communication, and AI night-vision methods. The expected result is a highly reliable traffic monitoring system that helps make better traffic light decisions. For society, this will reduce traffic jams and clear roads faster for emergency vehicles. Commercially, this system can be used as a smart and low-cost solution for city traffic management.

Keywords: traffic density detection, computer vision, Raspberry Pi, AI night vision, smart traffic control

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
CNN	Convolutional Neural Network
IoT	Internet of Things
ITS	Intelligent Transportation Systems
GPS	Global Positioning System
RGB	Red, Green, Blue (Color Image Format)
SDG	Sustainable Development Goals
ETA	Estimated Time of Arrival
API	Application Programming Interface
CPU	Central Processing Unit
GPU	Graphics Processing Unit
YOLO	You Only Look Once (Object Detection Model)
CCTV	Closed Circuit Television
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation

Table 1:LIST OF ABBREVIATIONS

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1. INTRODUCTION

1.1 Background and Context

Traffic congestion at intersections is a major problem in modern road transport systems. The problem becomes worse when traffic signals use fixed timing, because fixed systems cannot respond to the actual number of vehicles waiting on each road [3], [7]. At a four-way junction, one road may have a long queue while another road may have very little traffic, but a fixed timer [7] may still use almost the same signal pattern, resulting in inefficient traffic distribution, longer queues, and unnecessary waiting time.

This issue affects private drivers, public transport users, emergency responders, traffic police, and local communities. It is especially important in Sri Lanka because the country depends heavily on road transport. The National Road Master Plan states that the road sector carries approximately 95 percent of passenger traffic and 97 percent of freight traffic in Sri Lanka [1], highlighting the critical importance of efficient road infrastructure and junction performance. The same report also identifies congestion reduction and junction improvement as national transport priorities [1], [19].

The issue becomes more critical during emergencies. The 1990 Suwa Seriya ambulance service reports an average response time of 11 minutes and 32 seconds in Colombo and 14 minutes and 49 seconds island-wide [8]. It also uses GPS combined with traffic information to dispatch the fastest available ambulance rather than the nearest one [8], demonstrating that real-time traffic awareness is already an important component in emergency response systems. However, junction-level congestion can still delay movement if real-time traffic density is not accurately detected.

Existing traffic monitoring approaches include loop detectors, infrared sensors, radar, and cameras. Among these, camera-based systems are highly effective because a single camera can observe multiple vehicles simultaneously and provide data for counting, classification, and density estimation [3]. Recent studies show that camera-based machine learning systems and edge-deployed solutions using Raspberry Pi improve traffic monitoring and support adaptive traffic control [2], [4], [13], [14]. However, many existing systems still have limitations. Some require strong visual conditions, some focus only on emergency vehicle recognition, and others are not optimized for low-cost edge deployment. Additionally, many systems do not include a reliable fallback mechanism for poor visibility conditions such as low light, rain, or haze [5], [6], [17].

To address these issues, this research proposes a camera-based traffic density detection component for a four-way junction. Four cameras are deployed to monitor each incoming road. A Raspberry Pi processes live video using trained detection models to detect vehicles, count them, and estimate traffic density in real time. When visual performance is degraded, a thermal support approach is considered to improve reliability [6], [17]. The processed data are transmitted to the cloud and used by a Smart Emergency Traffic Light Control System for intelligent signal control and emergency route prioritization.

This research aligns with SDG 9 (Industry, Innovation and Infrastructure) and SDG 11 (Sustainable Cities and Communities) by supporting smarter transportation infrastructure and improved urban mobility. It also supports SDG 3 (Good Health and Well-being) because improved traffic awareness can help reduce emergency response delays [8].

1.2 Literature Review

Smart traffic management systems have become an important research area due to increasing urban congestion and the need for efficient transportation control. Modern intelligent transportation systems use computer vision, machine learning, and real-time sensing to monitor traffic conditions and support adaptive traffic signal control. Camera-based traffic monitoring is particularly important because it enables continuous observation of vehicle flow, density, and emergency vehicle movement at intersections.

Several research studies have explored traffic monitoring, vehicle detection, and adaptive signal control. These include camera-based density estimation, deep learning-based vehicle recognition, emergency vehicle prioritization, and intelligent traffic optimization methods. Reviewing these studies helps identify current capabilities and research limitations.

Camera-based traffic density estimation is a major research direction. Sangeetha et al. proposed a real-time smart traffic management system using video analysis and demonstrated that camera-based data can estimate traffic conditions effectively [2]. Paluba et al. also showed that YOLO-based video analysis can support automated traffic flow monitoring at intersections [4]. Edge-based implementations using Raspberry Pi further demonstrate the feasibility of low-cost density estimation systems [13], [14]. These studies confirm that camera-based systems are effective because they can monitor multiple lanes simultaneously.

Deep learning-based vehicle detection is another key area. Alfailakawi et al. demonstrated that YOLOv8 improves vehicle recognition in intelligent transportation systems [5]. Karoon et al. showed that vision-based deep learning can directly support adaptive traffic light control using density estimation [9]. EdgeML-based YOLO systems have also achieved real-time performance on Raspberry Pi devices for vehicle tracking and density calculation [15]. However, many of these studies focus on control algorithms rather than the design of a complete four-camera junction-level detection system.

Adaptive traffic control systems rely heavily on real-time traffic data. Alghamdi et al. developed a self-adaptive traffic signal system using machine learning-based optimization, showing improved performance compared to fixed timing systems [3]. Other studies confirm that adaptive traffic systems perform better when continuous density data is available [7], while IoT-based solutions support broader city-scale deployment [18]. However, these studies focus more on signal control rather than the underlying data acquisition component.

Emergency vehicle prioritization is another important research area. Vikraman et al. proposed a hybrid deep learning system to improve emergency response efficiency [10]. Khan et al. highlighted the importance of real-time traffic information for emergency signal prioritization [11]. YOLO-based emergency traffic systems also demonstrate strong results in detecting and prioritizing emergency vehicles [16]. However, these systems primarily focus on emergency classification rather than continuous multi-direction density estimation.

Visibility challenges remain a major limitation in camera-based systems. Bhadoriya et al. demonstrated that thermal imaging improves detection accuracy in low visibility conditions such as rain, fog, and darkness [6]. Recent reviews confirm that environmental conditions such as haze, occlusion, and lighting variations continue to affect system performance [12], [17]. This strongly supports the inclusion of fallback mechanisms in real-world systems.

In Sri Lanka, research in this area is still limited, but the practical need is clear. AI-based traffic studies highlight congestion and junction optimization as key priorities [19], while Suwa Seriya

demonstrates the importance of real-time traffic awareness in emergency response systems [8]. Therefore, there is a strong need for a practical, low-cost, and reliable junction-level traffic density detection system.

Overall, although camera-based traffic monitoring is well researched, limitations remain in terms of complete system design, low-cost deployment, and reliability under poor visibility conditions [2], [6], [9], [12], [17], [19].

1.3 Research Gap

The main research gap is that current studies do not fully provide a practical and affordable camera based traffic density detection component for a four way junction that continuously monitors all four road approaches and supports real time decision making. Many studies focus on general traffic analysis, emergency vehicle recognition, or adaptive signal control, but they do not clearly describe a complete four camera density monitoring system designed for a real junction with low cost edge processing [2],[5], [9],[11].

Another gap is reliability under poor visual conditions. Camera based systems are useful because they are flexible and information rich, but their performance can drop in low light, rain, haze, and other difficult conditions. Although thermal imaging research shows that thermal cameras can improve vehicle detection in adverse visibility conditions, many smart traffic density systems do not include a practical fallback or support approach for such cases [6], [12]. This leaves a reliability gap in real world deployment.

There is also a practical implementation gap. Some solutions are tested mainly in simulation or with stronger computing resources, while fewer studies focus on lightweight edge devices such as Raspberry Pi. For this project, edge processing is important because the system needs to be low cost, scalable, and suitable for practical deployment. This is especially relevant in a Sri Lankan setting where smart traffic solutions need to balance accuracy, speed, and cost [1], [2], [4].

This research addresses these gaps by proposing a camera based traffic density detection component that uses four cameras to observe a four way junction in real time, processes live video on a Raspberry Pi using trained detection models, estimates traffic density for each road direction, and sends the processed data to the cloud. In addition, the research considers a thermal support approach when visual detection is weak [6]. The novelty of this component is therefore the integration of four direction camera monitoring, real time density estimation, edge processing, and visibility support in one practical smart junction solution.

Therefore, the contribution of this research is not only vehicle detection. It is a practical traffic awareness component that helps the wider Smart Emergency Traffic Light Control System understand junction conditions in real time and make better decisions than fixed timer based systems [3], [7]

2. OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate a camera based traffic density detection component for a four way junction that can identify vehicles, count them, and estimate real time traffic density accurately enough to support intelligent traffic signal decisions, while also exploring a thermal support approach to improve vehicle detection reliability under poor visibility conditions such as low light, rain, fog, or haze.

2.2 Specific Objectives

1. To capture live traffic video from four cameras placed at the four incoming roads of the junction.
2. To develop a vehicle detection and counting method using computer vision models on a Raspberry Pi.
3. To estimate traffic density for each road direction in real time and send the processed results to the cloud.
4. To explore a thermal support approach for improving detection reliability under poor visibility conditions.
5. To evaluate the component using accuracy, processing speed, and communication performance measures.
6. To integrate the traffic density output with the wider Smart Emergency Traffic Light Control System

3. METHODOLOGY

Research Approach

This research uses System Development and Evaluation as the research methodology because the study focuses on designing, implementing, and testing a new technical component for the full system.

In this component, the work begins with requirement identification for camera placement, field of view, processing needs, and density estimation output. After that, four cameras will be selected and positioned to observe the four roads of the junction. Video samples will then be collected for model development and testing. The Raspberry Pi will be used as the edge device for local processing. Trained computer vision models will be used to detect vehicles, count them, and estimate traffic density.

The main technical methods in this component are video acquisition, image preprocessing, object detection, vehicle counting, traffic density estimation, edge processing, and cloud communication. The system will analyze incoming video streams and convert them into useful traffic information for each road. If the visual result becomes weak because of poor visibility, a thermal support approach will be considered.

The tools and platforms used in this component include surveillance cameras, Raspberry Pi, Python, OpenCV, deep learning based object detection models, and a cloud platform for sending processed traffic data. These tools are suitable because they support real time traffic observation and affordable edge implementation.

This is the main data collection component in the full project. Data collection for this member includes capturing junction video samples, preparing training and testing frames, labeling vehicles if necessary, and collecting output logs such as counts, density levels, and processing times. Ethical considerations include avoiding unnecessary personal identification, using the data only for traffic analysis, and ensuring responsible handling of recorded video.

Validation will be done using functional testing and quantitative performance evaluation. The main performance metrics will include vehicle detection accuracy, counting accuracy, density estimation accuracy, processing latency, and successful data transfer to the cloud. If thermal support is tested, its contribution to improved detection reliability under weak visibility will also be evaluated.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

System Architecture Diagram

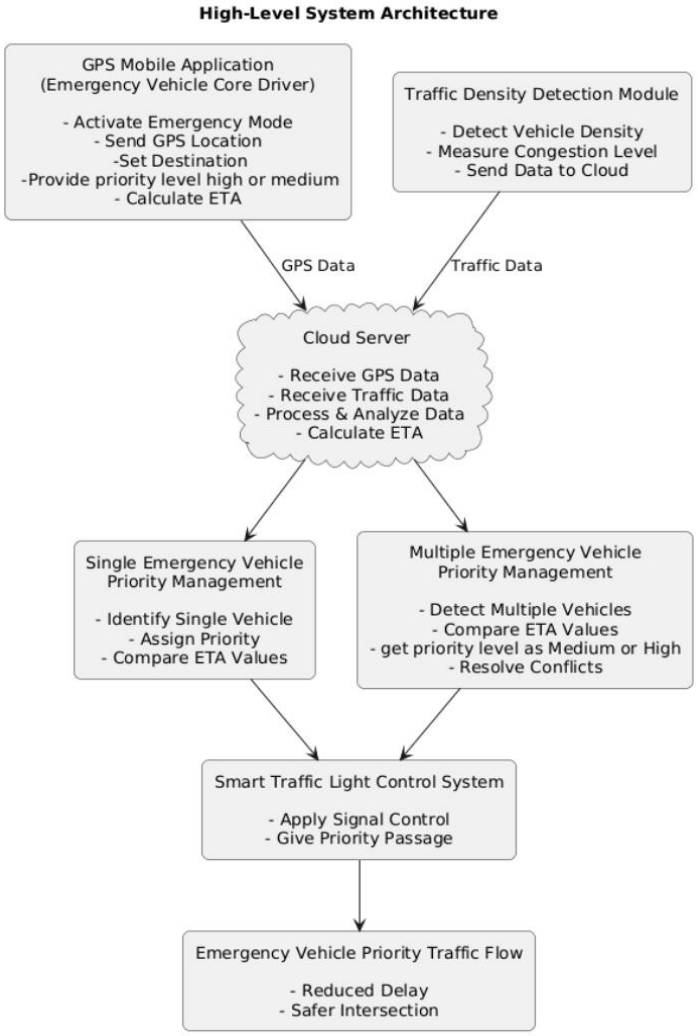


Figure 1: HIGH-LEVEL SYSTEM ARCHITECTURE

1. **GPS Mobile Application (Emergency Vehicle Side)** The process starts with the emergency vehicle driver. The driver uses a specially designed mobile app and activates the "Emergency ON" mode. Instantly, the app sends the vehicle's live GPS location, current speed, and Estimated Time of Arrival (ETA) directly to the cloud server.

2. **Traffic Density Detection (Junction Side)** While the app sends vehicle data, four cameras at the traffic junction monitor the real-time road conditions. Using a Raspberry Pi 4, the system detects and counts the vehicles to measure the traffic jam. If the visual is unclear due to bad weather or darkness, thermal cameras are used as a backup. This traffic density data is also sent to the cloud.

3. **Cloud Server (The Central Brain)** The cloud server acts as the main connection hub for the entire system. It safely receives, stores, and processes the live GPS data from the mobile app

and the traffic congestion data from the cameras. It combines all this information to help the system make the right decisions.

4. **Single Emergency Vehicle Priority Management** When the cloud detects that only **one** emergency vehicle is approaching the junction, this specific module takes charge. It tracks the vehicle's location, calculates the timing, and prepares to give it the highest priority for a clear and safe path.

5. **Multiple Emergency Vehicle Priority Management** If **two or more** emergency vehicles approach the same junction from different directions at the exact same time, this separate module is activated. It compares their distance, ETA, and urgency levels (for example, giving first priority to a critical ambulance over a fire truck) to smartly decide which vehicle should cross the junction first.

6. **Smart Traffic Light Control & Alert Buzzer (Taking Action)** Based on the priority decisions from the modules above, the **Smart Traffic Light Control** automatically changes the traffic lights to green for the selected emergency vehicle. If the traffic jam is too heavy and simply changing lights is not enough, the **Emergency Alert Buzzer** is activated to warn nearby drivers to quickly move out of the way.

7. **Fail-Safe Mechanism (System Safety)** To ensure maximum safety, the system has a built-in failsafe feature. If the system detects any technical error, hardware failure, or cyberattack, it automatically switches all traffic lights to "Yellow Blinking" mode. This keeps the junction safe and operates like a normal manual intersection until the issue is fixed.

Component Diagram

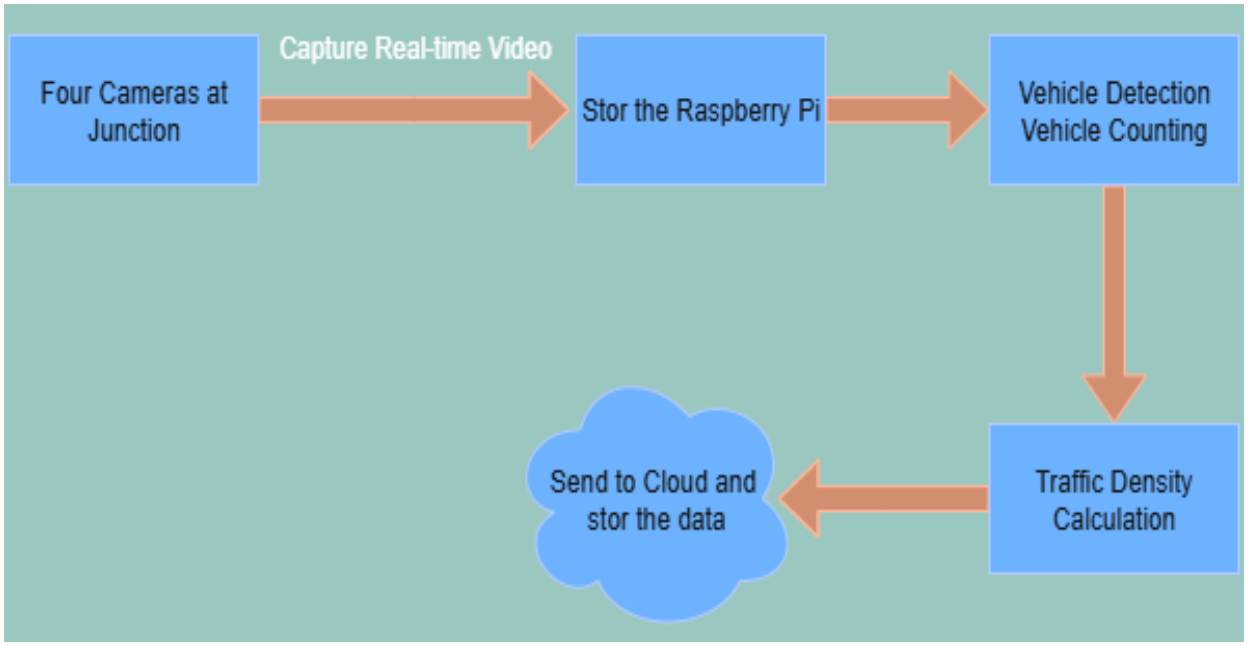


Figure 2: Component Diagram

The proposed system architecture describes how the camera-based traffic density detection component monitors and analyses traffic conditions at a four-way junction. In this approach, four surveillance cameras are installed to observe the four incoming roads of the junction. These cameras continuously capture live traffic video streams and provide the input data required for traffic analysis. The video streams

are transferred to the processing system through the communication layer, where frames are extracted and prepared for further analysis.

In the processing layer, the system performs several operations to convert the captured video into useful traffic information. First, the image preprocessing module enhances the captured frames to improve image quality. After that, the vehicle detection module uses computer vision models to detect vehicles from the video frames and count them. Based on the detected vehicles in each road direction, the traffic density analysis module estimates the congestion level at the junction. The final processed traffic density data is then sent to a cloud database or cloud server where it can be used by the wider Smart Emergency Traffic Light Control System.

This architecture also shows how different modules interact and how data flows through the system. Video data flows from the cameras to the frame capture module, then through preprocessing and detection modules, and finally to the traffic density analysis module. The system is designed to be scalable because additional cameras or junctions can be added using the same architecture. Security and privacy are also considered by transmitting only processed traffic information to the cloud instead of raw video, reducing unnecessary exposure of sensitive data.

5. USER REQUIREMENTS

The main stakeholders for this component are traffic authorities, emergency response teams, road users, system administrators, and municipal planners. The component also indirectly benefits patients, hospitals, and the general public because better traffic density awareness supports faster emergency route management.

Functional Requirements

1. The system shall capture live video from four cameras placed at the four roads of the junction.
2. The system shall detect vehicles from the incoming video streams.
3. The system shall count vehicles for each road direction.
4. The system shall estimate traffic density for each approach road in real time.
5. The system shall send processed traffic density data to the cloud.
6. The system shall support integration with the wider smart traffic signal control system.
7. The system shall support an optional thermal based method when visual detection becomes unclear.

Non Functional Requirements

1. Security: The system shall ensure secure communication between the edge device and the cloud platform.
2. Performance: The system shall process traffic video with acceptable delay for real time use.
3. Accuracy: The system shall provide sufficiently accurate vehicle counts and density estimates for traffic decision support.
4. Usability: The component shall be easy to configure and monitor by system operators.
5. Scalability: The design shall support future expansion to more junctions

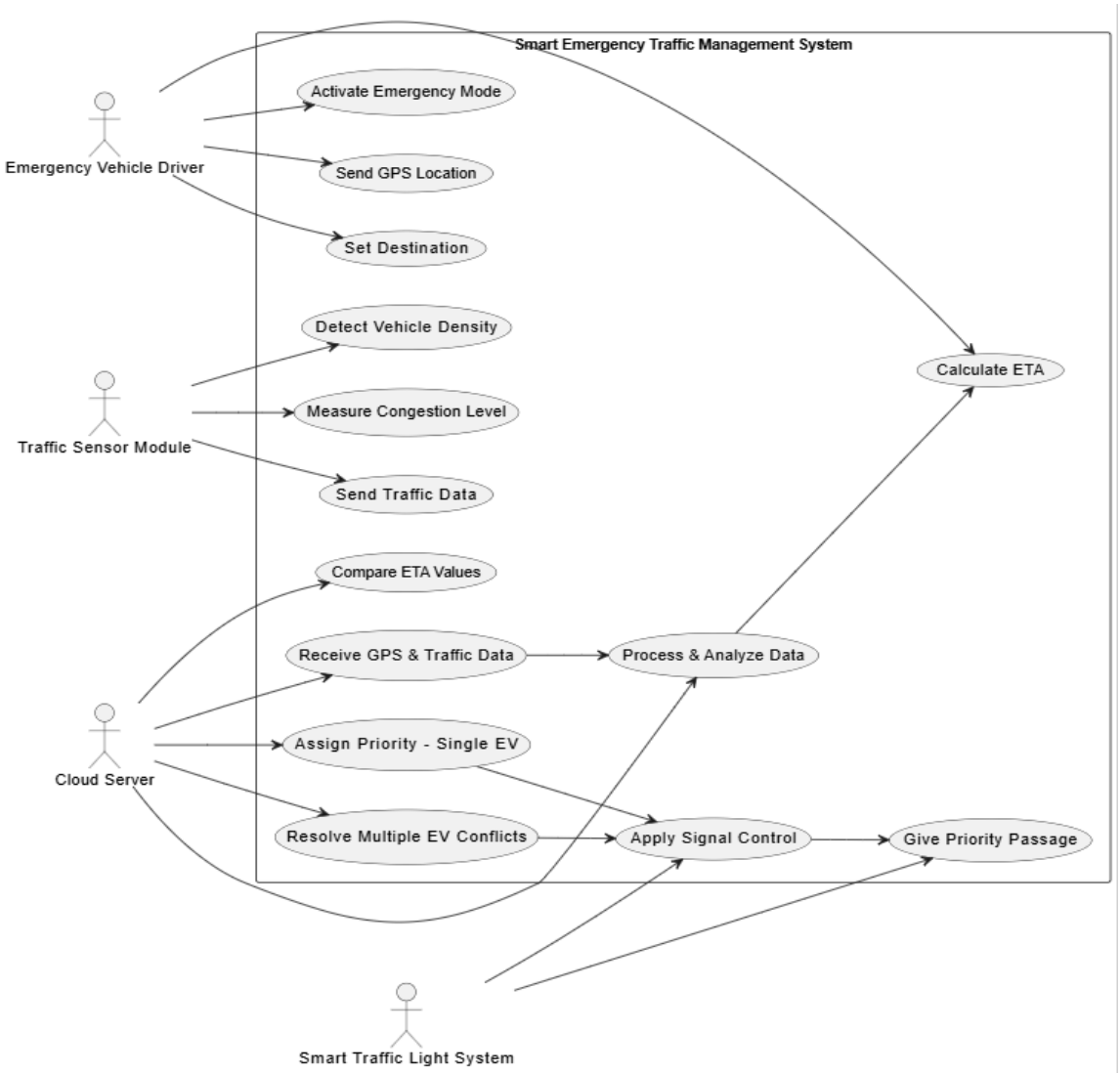


Figure 3: use case diagram

6. COMMERCIALIZATION PLAN

- The target market for this component includes municipal councils, urban traffic authorities, smart city platform providers, and emergency traffic management services. In Sri Lanka, this solution can be useful in busy urban junctions where congestion and emergency delays are common.

- The main value proposition is that this component offers a practical and relatively low cost method for monitoring real traffic density at a four way junction using cameras and edge computing. Unlike fixed timer systems, it provides live traffic information that can support smarter and more responsive signal control.

- The revenue model can be based on institutional deployment and maintenance. Municipal authorities or smart city service providers can pay for installation, software customization, monitoring support, and annual maintenance. Additional revenue can come from scaling the system to multiple junctions.

- The pricing strategy can be based on the number of junctions, the number of cameras, and the chosen service package. The competitive advantage is that this solution is practical, edge based, and more affordable than some infrastructure heavy monitoring systems. Intellectual property can be considered for the software logic, edge processing workflow, and system integration approach.

7. BUDGET AND JUSTIFICATION

The overall estimated budget for the Smart Emergency Traffic Light Control System is LKR 233,750. This budget covers the main hardware, cloud services, API services, and prototype construction materials required for the implementation and testing of the complete system.

Item	Qty	Estimated Cost (LKR)	Justification
Raspberry Pi 4 / 5 Board (8GB RAM)	1	80,000	To analyze video data from AI cameras and run traffic algorithms at high speed.
NodeMCU (ESP32 / ESP8266) Boards	As needed	2,750	To receive signals from the cloud and control intersection LED traffic lights.
High-Definition Web Cameras	4 units	13,500	Live detection of vehicle traffic across 4 intersection roads for computer vision.
4x4 Prototype Board & Base Materials	1 set	4,500	Wood or acrylic boards to construct the physical intersection prototype.
PVC Pipes & Fittings	As needed	3,000	To build the prototype poles resembling real traffic light poles.
LED Lights, Relays, Wires & Breadboards	As needed	12,000	For red, yellow, green indicator lights and circuit connections.
Power Supplies (5V & 12V Adapters)	As needed	8,000	To provide uninterrupted power to the Raspberry Pi and NodeMCU boards.
Firebase Cloud Services (Blaze Plan)	1 Plan	45,000	Real-time database for low-latency data transfer between the App, AI cameras, and Hardware.
Google Maps & Directions APIs	Premium	65,000	For accurate GPS location mapping, distance calculations, and route planning.
Total Estimated Cost			**233,750**

Table4 :BUDGET AND JUSTIFICATION

8. WORK BREAKDOWN STRUCTURE (WBS)

Project Phase	Timeline	Primary Activities	Milestones / Deliverables
Phase 1: Proposal Development and Requirement Analysis	Months 1 to 2	Conduct literature review, identify the research problem and system requirements, define camera placement and processing needs, and prepare proposal materials.	Finalized proposal and completed requirement analysis.
Phase 2: System Design and Simulation Planning	Months 3 to 4	Design the four camera monitoring setup, plan the data collection process, define the traffic density estimation logic, and prepare the system architecture.	Completed system design and data collection plan.
Phase 3: Core Development and IoT Integration	Months 5 to 6	Implement vehicle detection, vehicle counting, and traffic density estimation on Raspberry Pi, and connect the component to the cloud platform.	Completed core camera-based density detection component.
Phase 4: Prototype Implementation and Security Enhancement	Month 7	Implement the working prototype, improve system performance, and integrate thermal support for difficult visibility conditions if required.	Completed real prototype.
Phase 5: Testing, Validation, and Finalization	Months 8 to 9	Test detection accuracy, processing latency, and cloud communication, correct system errors, and finalize documentation.	Completed validation report and final Member 2 documentation.

Table 5:WORK BREAKDOWN STRUCTURE (WBS)

9. GANTT CHART

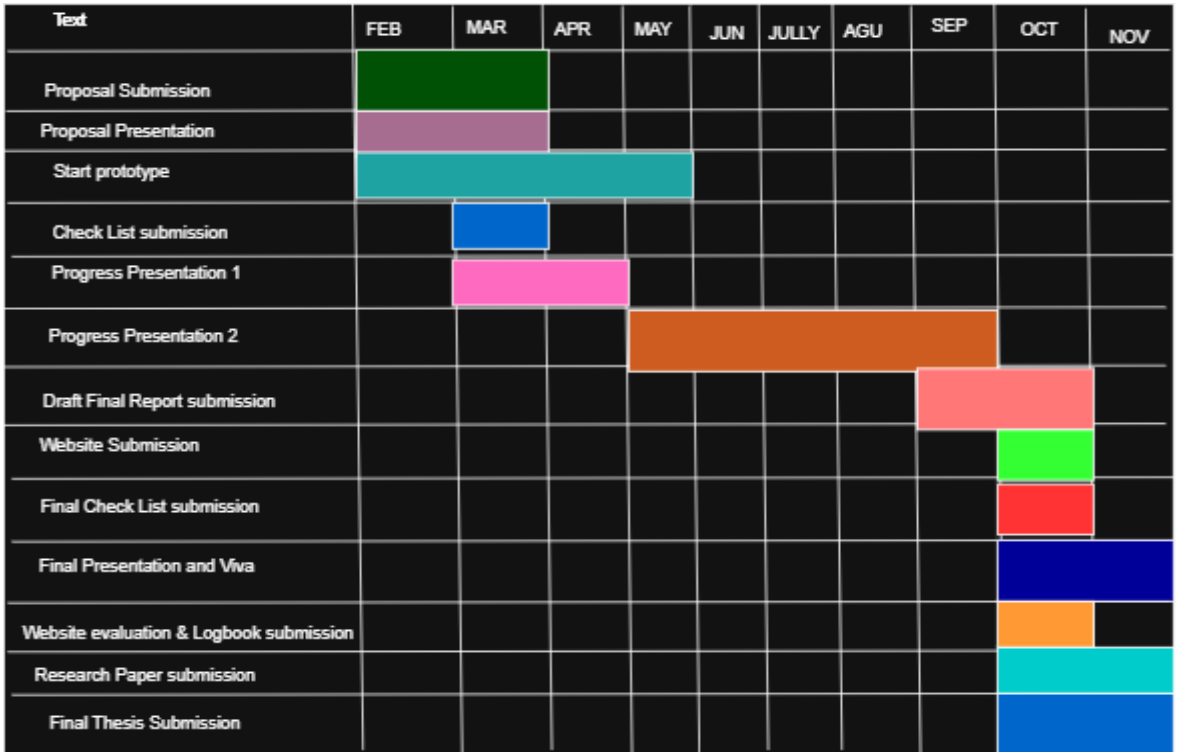


Figure 4: GANTT CHART 1

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11.APPENDICES

- Detailed diagrams
- Ethical Document

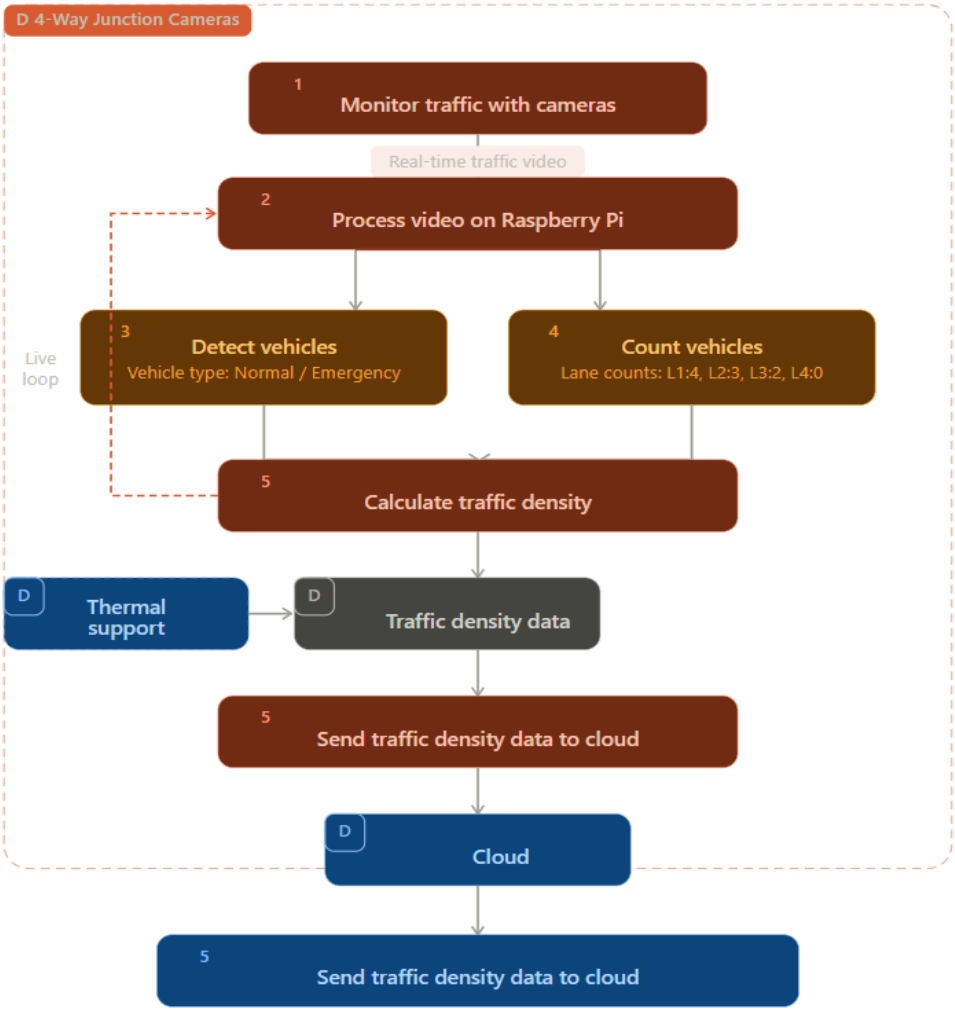


Figure 5: APPENDICES

Appendix 1– Camera Calibration and System Validation Activities

This appendix summarizes the calibration and validation procedures used to ensure accurate traffic data collection before system deployment. Proper calibration improves the reliability of vehicle detection and traffic density estimation models.

Category	Item Description	Present Study	Reference Study
Camera Calibration & System Validation	Calibrate camera angle and positioning for full road coverage	0.68	0.72
	Adjust brightness and contrast for different lighting conditions	0.65	0.70
	Validate vehicle detection accuracy using labeled datasets	0.74	0.78
	Test system performance under different weather conditions (rain, low light)	0.60	0.69
	Verify Raspberry Pi processing accuracy for real-time detection	0.66	0.73

Table 2 : Camera Calibration and Validation Procedures and Performance Comparison Between the Present Study and Reference Study

Appendix 2 – Traffic Density Assessment Metrics

This appendix describes how traffic parameters are analyzed to determine traffic conditions and support intelligent signal control decisions.

Category	Item Description	Present Study	Reference Study
Traffic Density Assessment	Count number of vehicles per lane using camera input	0.72	0.75
	Classify traffic density levels (low, medium, high)	0.67	0.70
	Measure queue length at each junction approach	0.64	0.68
	Analyze real-time traffic flow variation	0.66	0.71
	Evaluate congestion level for adaptive signal timing	0.69	0.74

Table 3 Traffic Density Assessment Metrics and Performance Comparison Between the Present Study and Reference Study

Appendix 3 – Machine Learning Model Development and Evaluation

This appendix summarizes the machine learning approaches used for training and evaluating models for vehicle detection and traffic density estimation.

Category	Item Description	Present Study	Reference Study
Data Analysis & ML Modeling	Train YOLO-based model for vehicle detection	0.72	0.78
	Train model for traffic density classification (low, medium, high)	0.68	0.73
	Train model for vehicle counting and tracking	0.70	0.75
	Evaluate model performance using cross-validation	0.74	0.79

Table6 : Performance Comparison of Machine Learning Models for Traffic Density Detection Between the Present Study and Reference Study

appendix 4 – System Automation and IoT Communication

This appendix outlines the automation mechanisms and IoT communication features implemented in the traffic monitoring system.

Category	Item Description	Present Study	Reference Study
System Planning & Automation	Develop automated traffic signal adjustment based on density data	0.69	0.75
	Implement alert system for emergency vehicle detection and priority	0.64	0.70
	Optimize IoT communication between Raspberry Pi and cloud server	0.71	0.76
	Enable real-time data transmission for traffic monitoring dashboard	0.68	0.73

Table 7: Comparison of System Planning and Automation Performance in the Present Study and Reference Study